

OBJECT MODELING ASSIGNMENT

Class Diagram, Object Diagram and Sequence Diagram

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Assignment	Class Diagram, Object Diagram and Sequence Diagram
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1. School Results Application – Class Diagram

Question

The class diagram represents the structure of a school results application where students have subjects, and their scores are calculated for grades.

Diagram Description:

- Classes: Student, Subject, GradeCalculator
- Relationships:
 - A Student has multiple Subject entries (Aggregation).
 - GradeCalculator computes the results for a Student.

→ Draw the Class Diagram

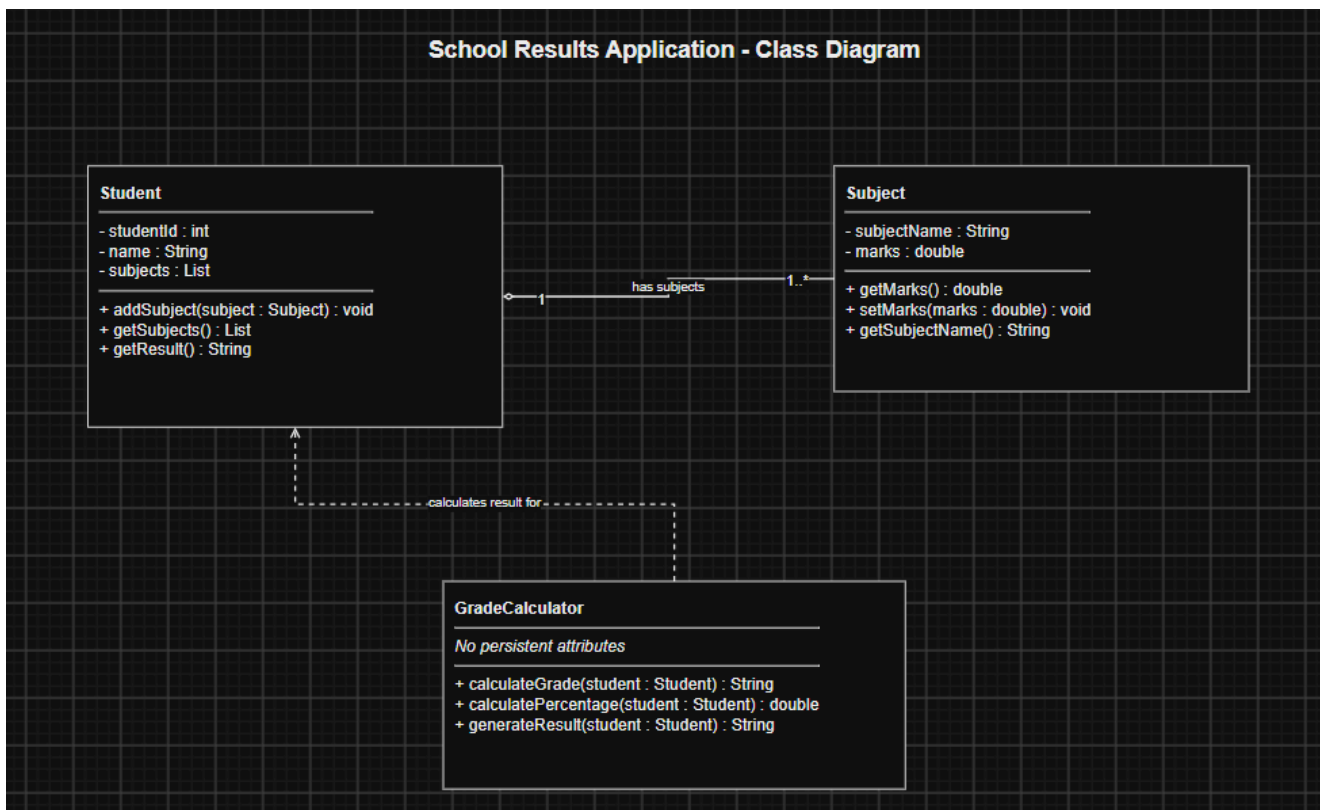


Figure 1: School Results Application – Class Diagram

2. School Results Application – Object Diagram

Question

An object diagram provides a snapshot of the Student and their Subject objects at a particular point.

Example:

- Student: John
- Subjects: Maths, Science
- Marks: 90, 85

→ Draw the Object Diagram

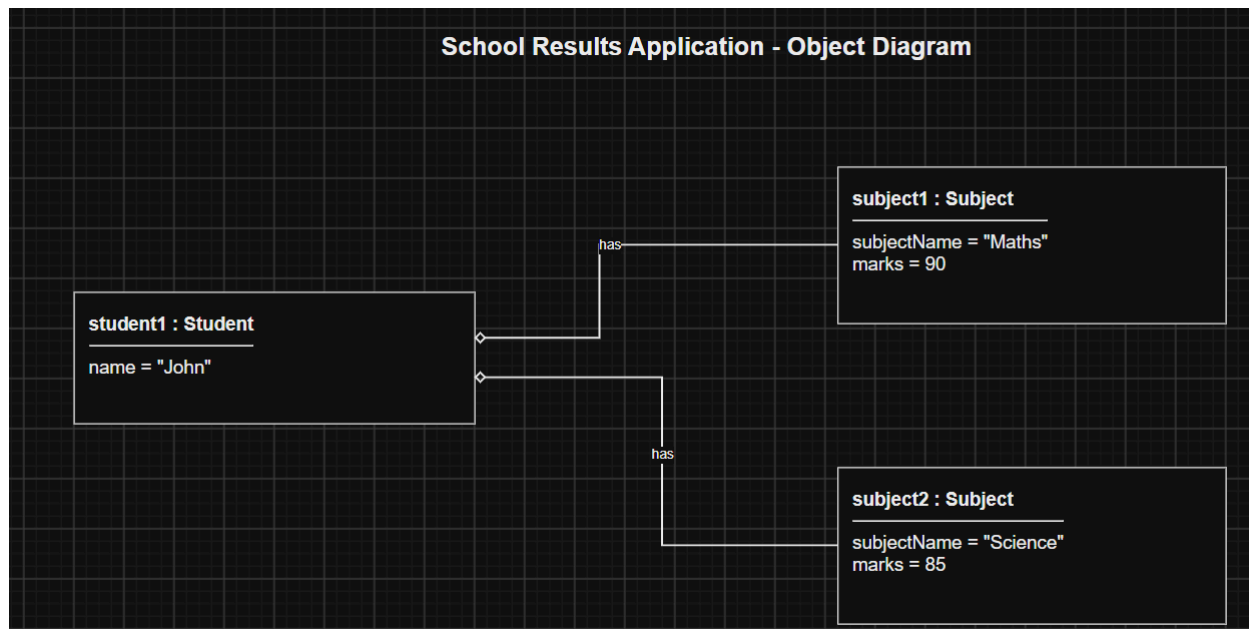


Figure 2: School Results Application – Object Diagram

3. School Results Application – Sequence Diagram

Question

The sequence diagram shows how objects interact to calculate grades.

Scenario: A student requests their grade based on marks in subjects.

Actors:

- 1. Student
- 2. GradeCalculator

→ Draw the Sequence Diagram

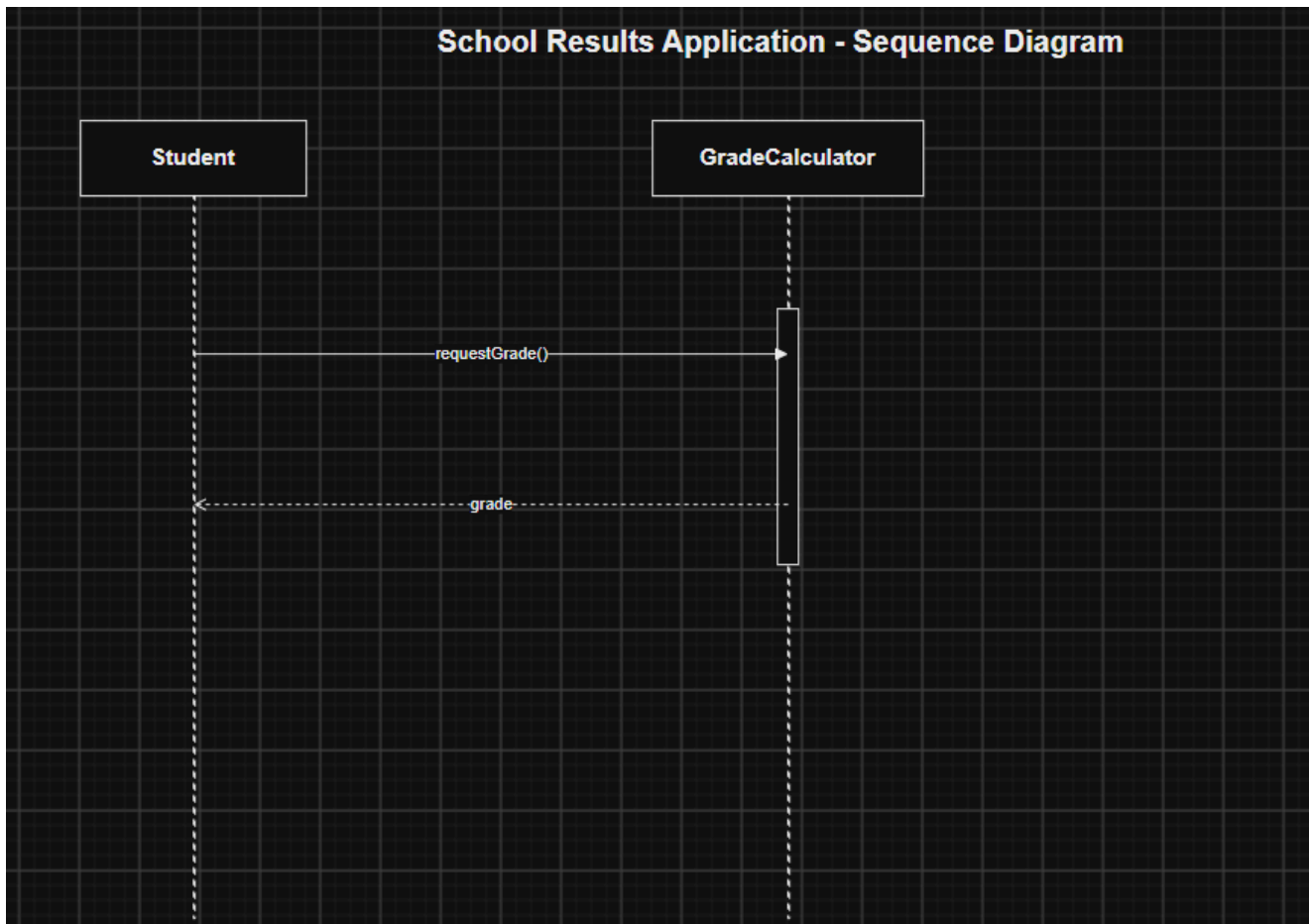


Figure 3: School Results Application – Sequence Diagram

4. Grocery Store Bill Generation Application – Class Diagram

Question

The class diagram models the system where a customer buys products, and the bill is generated.

Diagram Description:

- Classes: Customer, Product, BillGenerator
- Relationships:
 - A Customer can purchase multiple Product items (Composition).
 - BillGenerator computes the total for the Customer.

→ Draw the Class Diagram

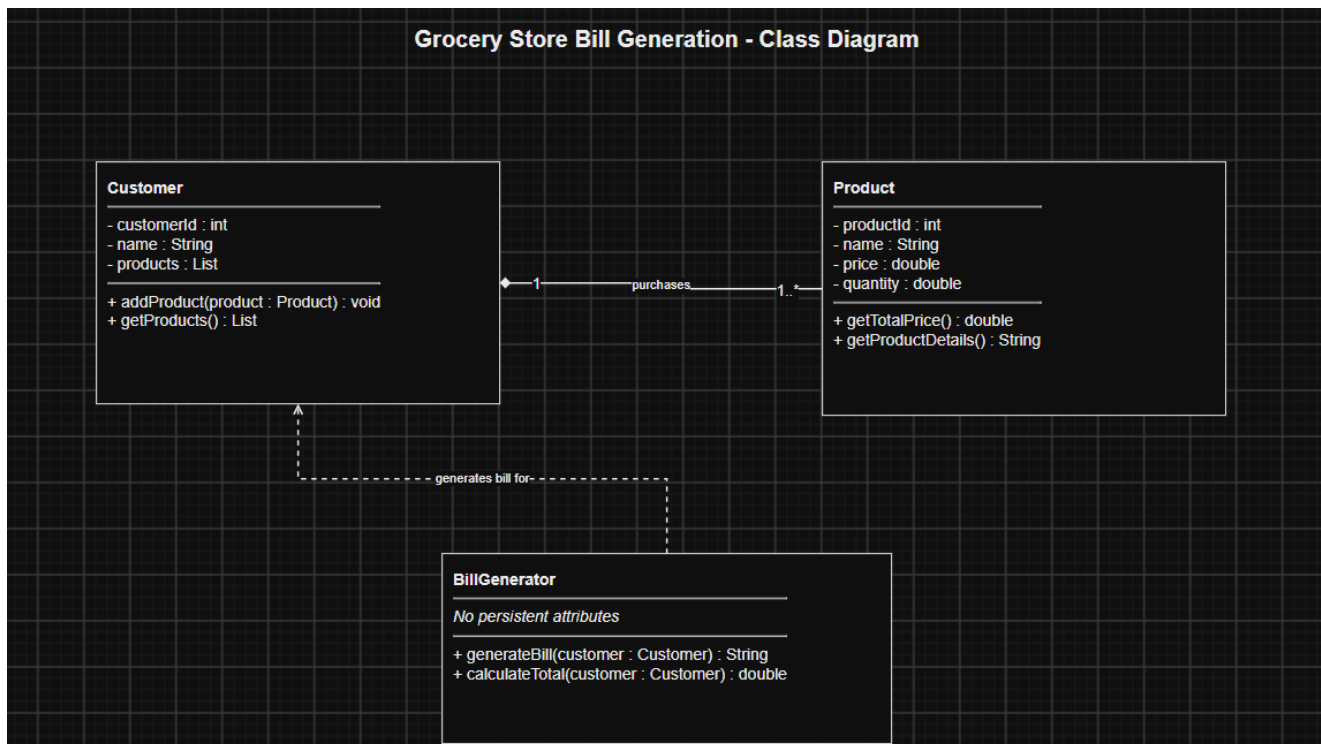


Figure 4: Grocery Store Bill Generation – Class Diagram

5. Grocery Store Bill Generation Application – Object Diagram

Question

An object diagram shows the details of a Customer and the Product objects they have purchased.

Example:

- Customer: Alice
- Products:
 - Apples (2 kg at \$3 per kg)
 - Milk (1 liter at \$2 per liter)

→ Draw the Object Diagram

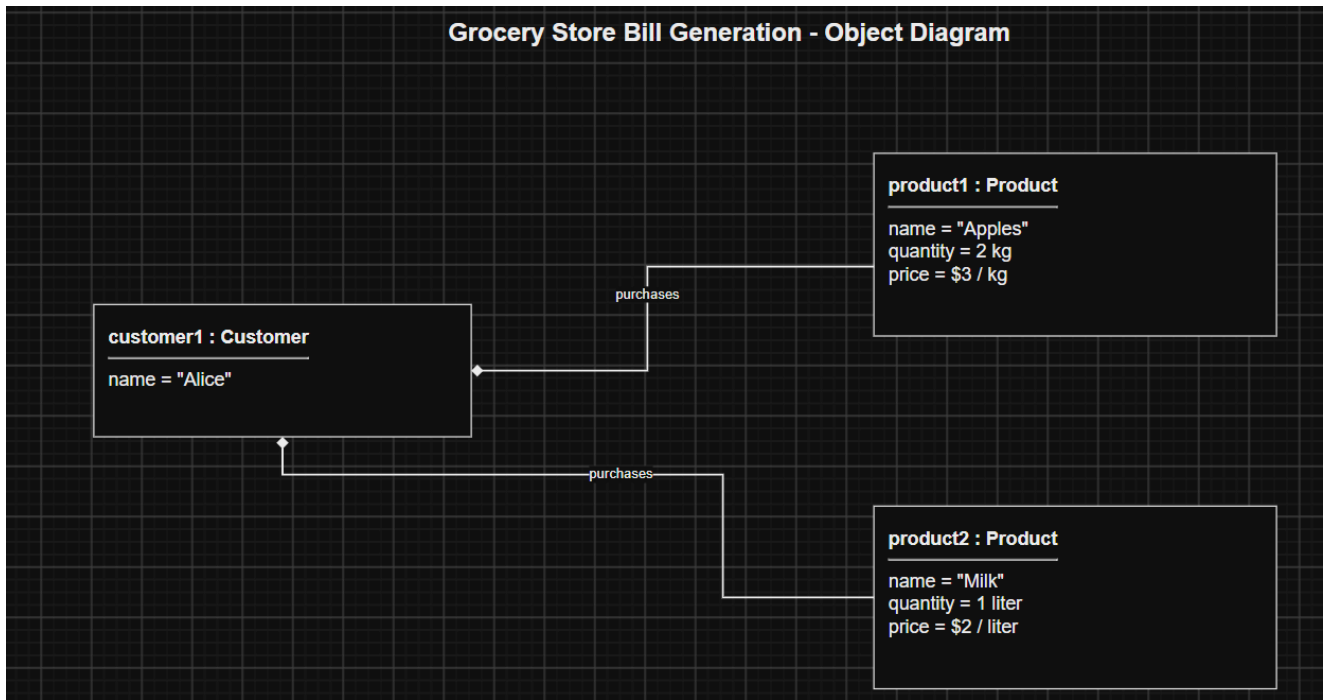


Figure 5: Grocery Store Bill Generation – Object Diagram

6. Grocery Store Bill Generation Application – Sequence Diagram

Question

The sequence diagram shows the process of bill generation for a customer.

Scenario: A customer checks out at the grocery store, and the total bill is generated.

Actors:

- 1. Customer
- 2. BillGenerator

→ Draw the Sequence Diagram

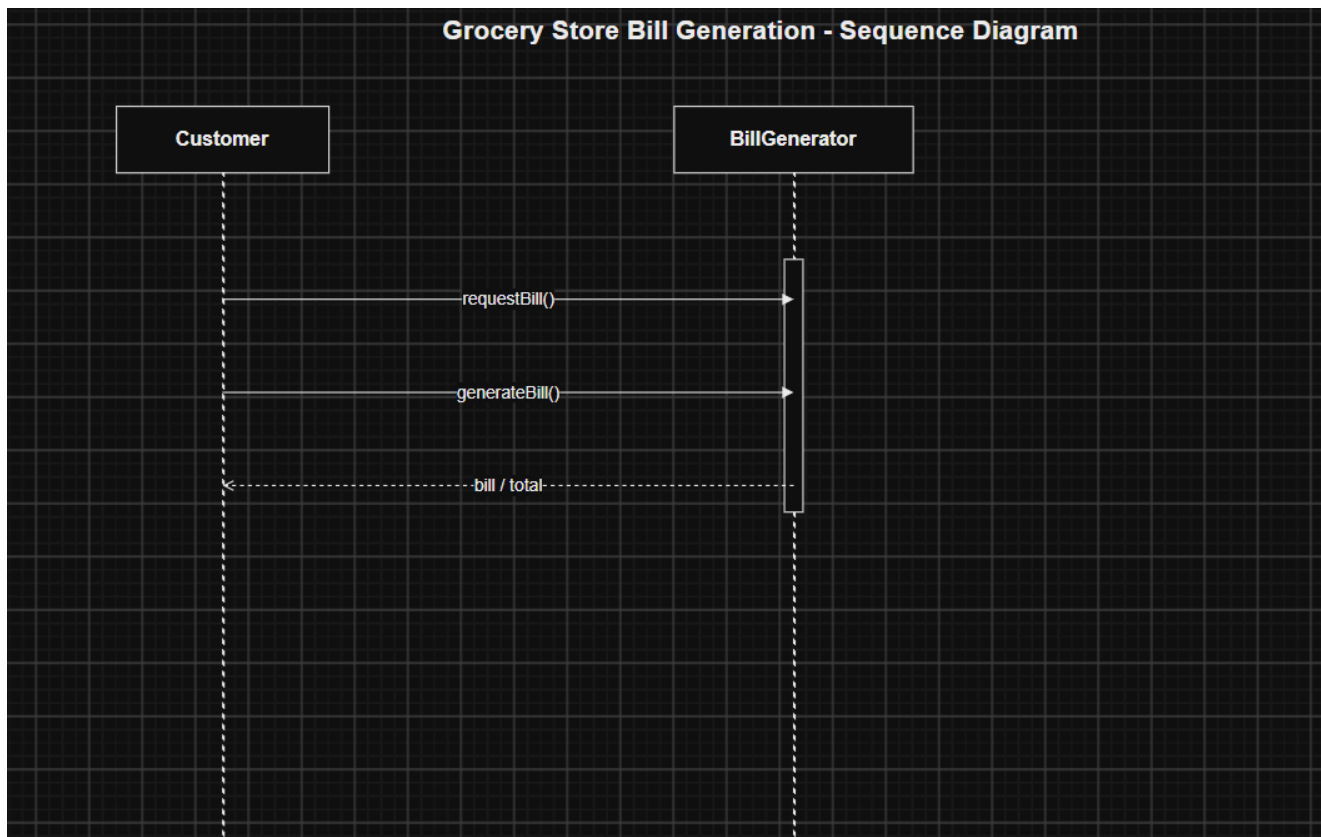


Figure 6: Grocery Store Bill Generation – Sequence Diagram